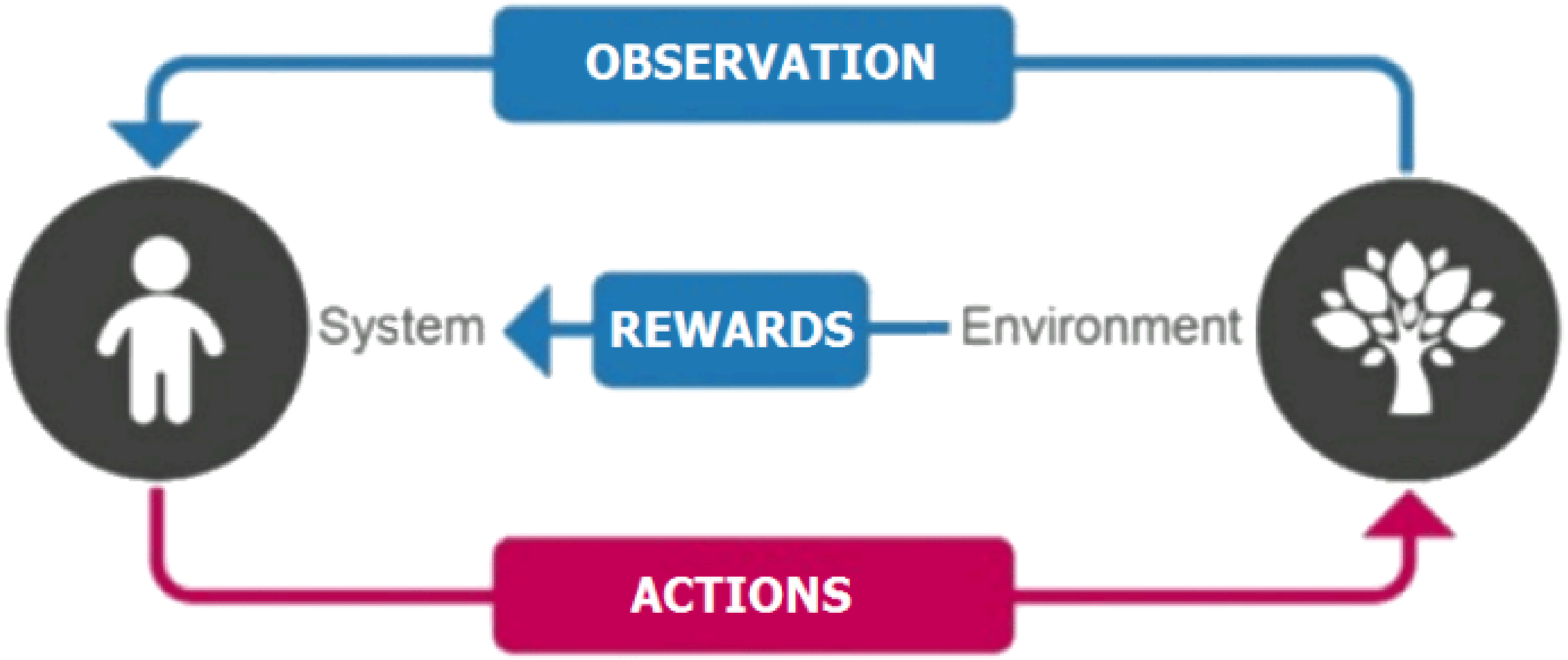
By: Manish

Introduction to AGENT

An LLM agent is an AI system that uses a large language model (LLM) to perform tasks like reasoning, planning, and decision-making.



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Prompt Engineering

It is the process of writing instructions to get the best results from an LLM models.

Prompting Techniques

1. Zero-Shot prompting
2. Few-Shot prompting
3. Chain-of-Thought (CoT)
4. Role Prompting

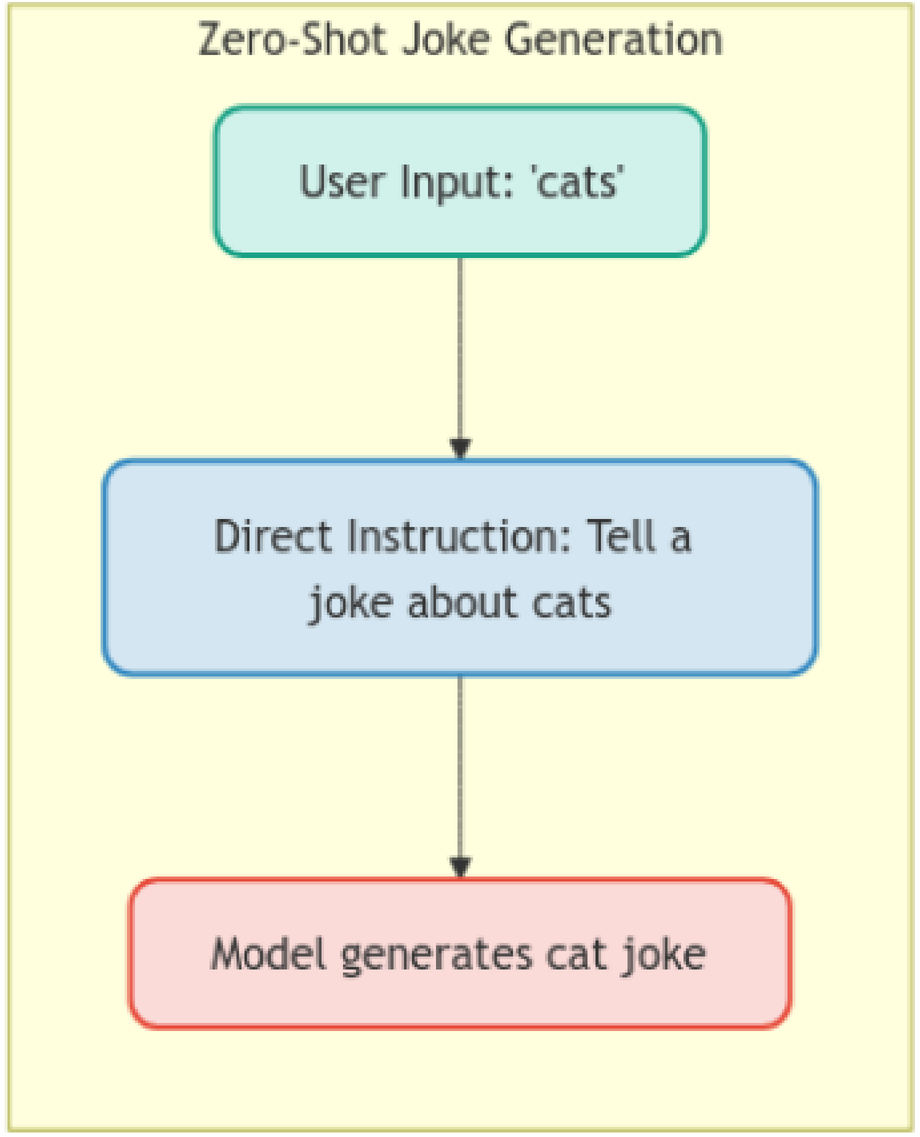
Prompt Engineering

1. Zero-shot Prompting

Prompting without any examples or demonstrations

Example Prompt:

Tell a joke about {animal}.



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Prompt Engineering

2. Few-shot Prompting

Prompting with some examples or demonstrations

Example Prompt:

Here are some animal jokes to learn from:

1. Why don't dogs make good comedians?
Their tales are too "ruff"!

2. What do you call a bird that's afraid of heights?
A chicken!

Using these examples as inspiration, create a similar-style joke about {animal}.

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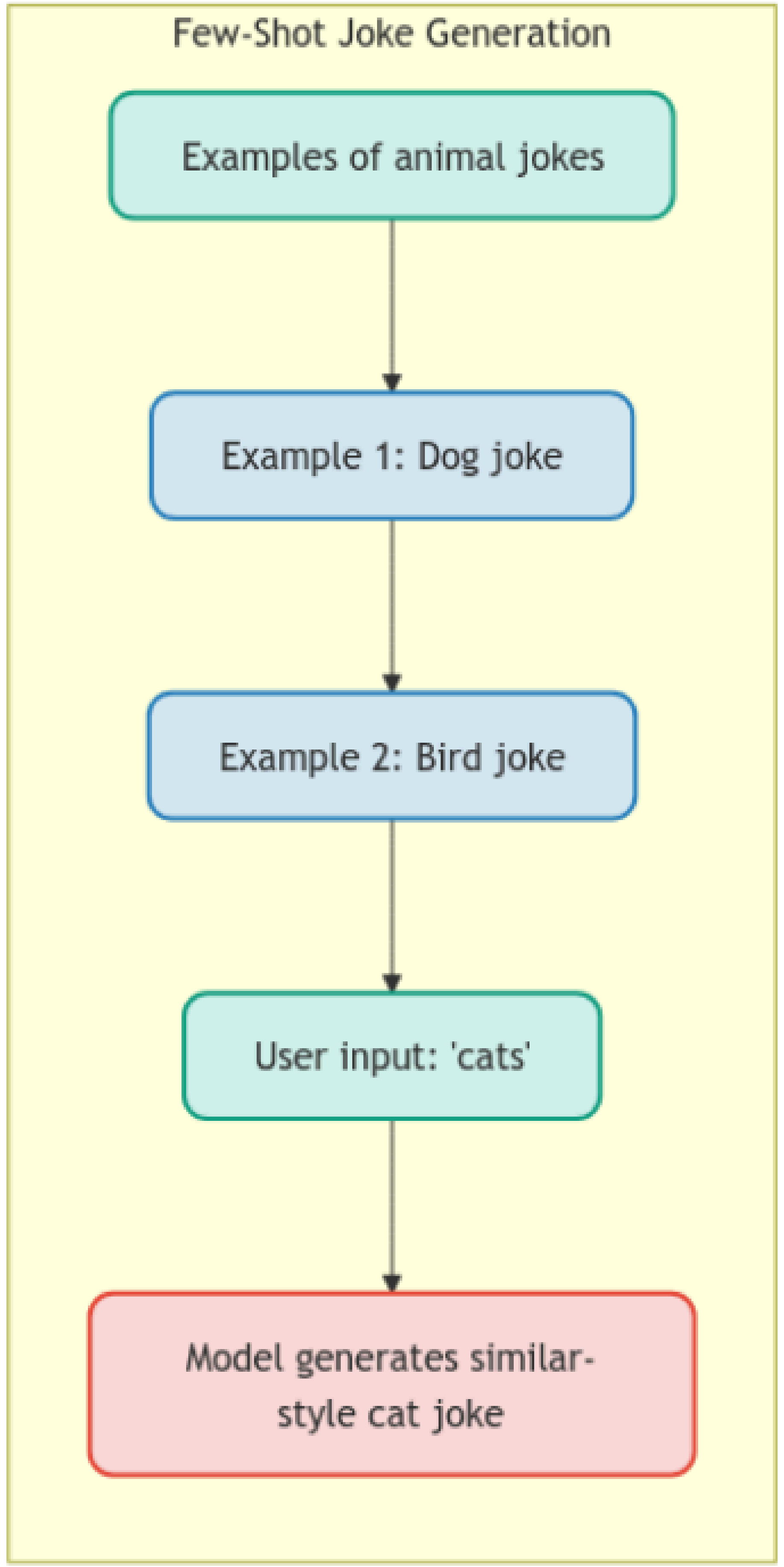

Prompt Engineering

3. CoT Prompting

Step by step instructions and breaking down the writing process .

Example Prompt:

Let's create a joke about {animal}

Think step by step:

1. First, identify key characteristics of given animal
2. Let's find wordplay opportunities with given animal or related terms. 3. Create a setup for the joke
4. Craft the punchline using our wordplay

Please follow these steps to construct a joke and response final answer.

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Prompt Engineering

4. Role Prompting

Providing the specific role and instructions before asking the questions,encouraging to get output from this role’s viewpoint.

Example Prompt:

You are a professional comedian who specializes in animal-themed comedy. You've been performing at comedy clubs for 15 years and have mastered the art of animal jokes. You understand animals behaviour deeply and know how to craft jokes that resonate with both animal owners and general audiences.

Your style is:

- Clean humour (family-friendly)
- Clever wordplay
- Observational comedy about typical animal behaviours
- Timing and setup are crucial elements of your jokes

As an experienced animal comedy specialist, please create a joke about {animal}. Include your thought process about why this joke works and how it plays on common animal behaviours or characteristics.

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Best Practices in Prompting

1. Be Specific
2. Use delimiters
3. Iterate: Test and refine prompt

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2. Use delimiters like :

<Instructions>

Joke should be full of punchlines.

</Instructions>

<steps>

Analyze the topic of the joke

1. Don’t ask like “Tell me a joke”. Instead add some more detail like : “Tell me a joke about current issue of road constructions”

Find the relevant context for the joke

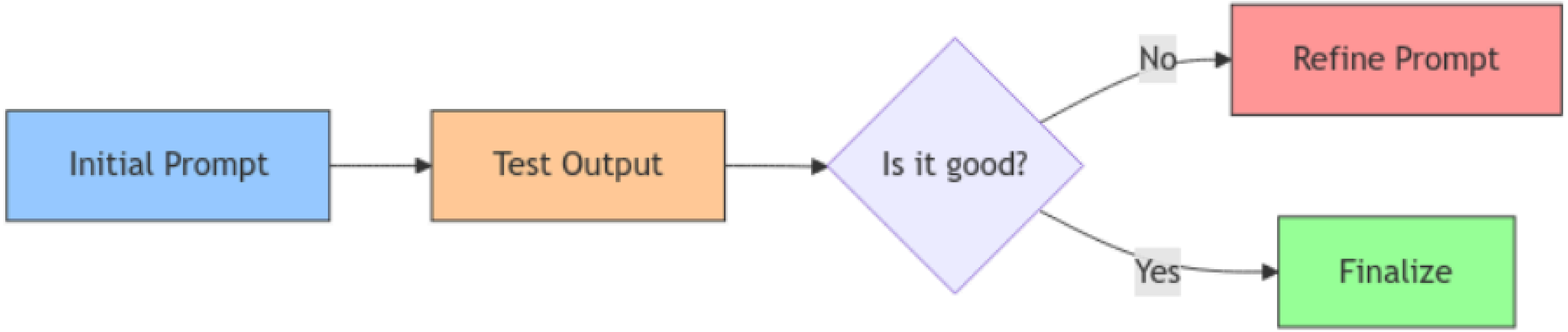
Create an setup

Add punchlines

</step>

Joke should be in below tag of response.

<response>...</response>



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Prompt Engineering

Prompt engineering is the heart of the LLM models

Better the prompt better the outcome

References for prompt engineering

https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for developers/

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Making LLMs More Deterministic

Causes of non-deterministic responses:

- LLMs generate responses through probabilistic sampling, where each word (token) is selected based on calculated probability distributions rather than fixed rules.

Ways to make more deterministic

- By passing low temperature in LLM requests
- By using more detailed instructed prompt

Why Non-deterministic is a problem

- Can be problematic for consistent business applications.
- Makes testing and validation more challenging.
- May produce inconsistent result in critical applications.

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Structured Output

Structured Response is the most needed feature for every developer right now .

In Early stage , developer have to write specific prompt to get desired response.

To overcome this type of unstructured response format , many frameworks have provided the structured output feature.

LLM Agent Workflow

LLM Agents combine language models with planning, memory, and tools to perform complex tasks

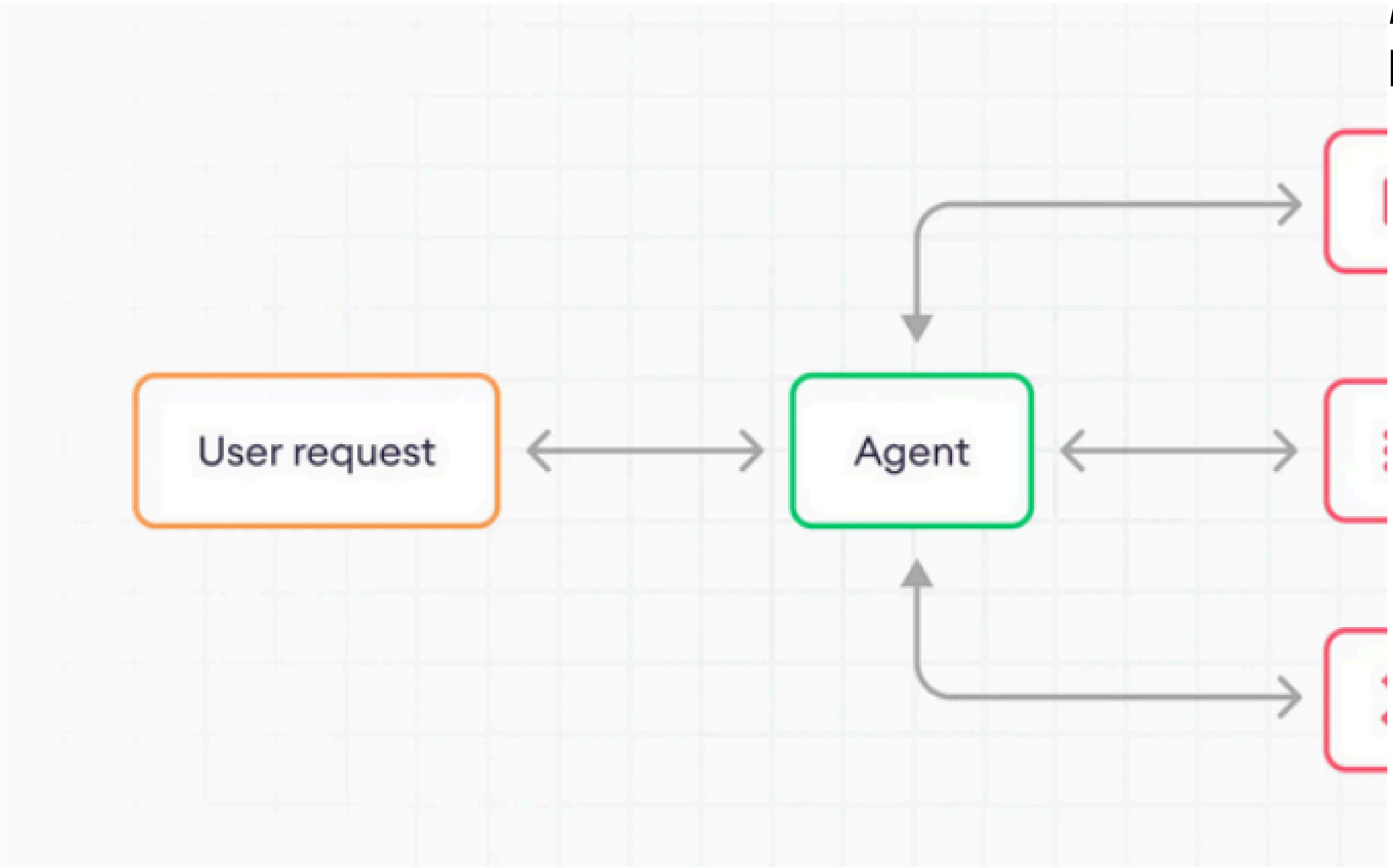

Fig: LLM Agent Workflow
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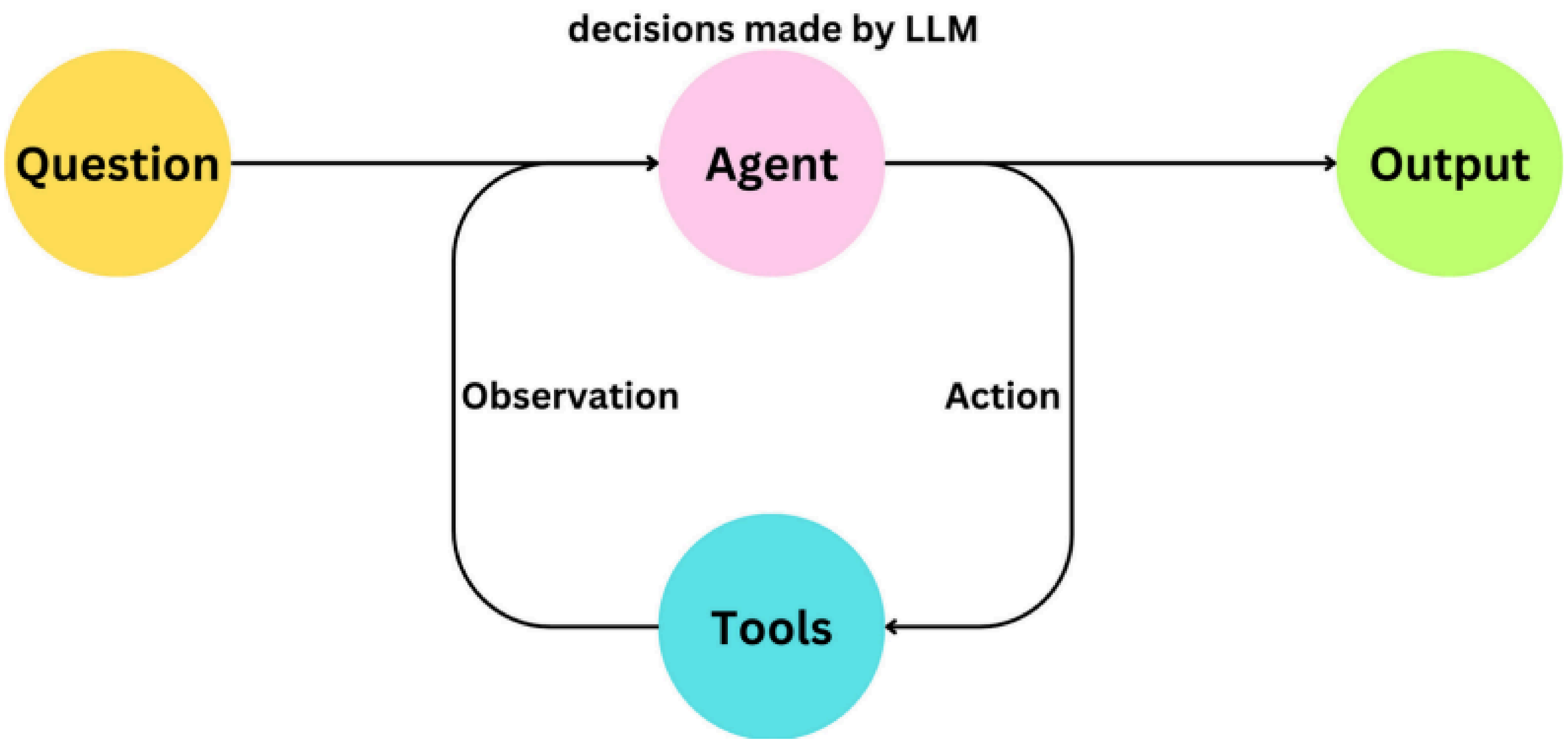
Tools

Tools are the functions that are designed to perform certain task.

Examples of tools : Web Search, Calculator , API calls, File Operation

When to use??

When the agent needs to perform specific tasks beyond text generations.



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Planning

Planning is used for reasoning and breaking down complex tasks into smaller, manageable steps.

a. Prompt Templates

Structure the input to ensure consistent agent behavior.

Help maintain consistent formatting and instruction patterns. b.

Chain of Thought

Breaks down complex tasks into simpler steps.

Supports multi-step reasoning and complex problem-solving.

Memory

Short term memory

It maintains the conversation history for the execution of time and for n number of conversations.

Think of it like a whiteboard.

Long-term memory

Its like a diary ,storing the insights and information from the past interactions . Good to use the vector database for this type of memory.

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Knowledge Base

A knowledge base is structured or semi structured information that an agent can search, to improve its responses.

The knowledge base can take different forms, such as:

- Static documents : PDFs,FAQs,...etc
- Vector databases: Embedding based retrieval system
- Databases : SQL or NoSQL databases

Graph Databases: Knowledge graphs like Neo4j

Vector Databases

Vector database offers the best solution for retrieving the relevant results from dense information quickly.

Steps Involved in AI Agent

1. Chunk the information
2. Generating Embedding and Storing to Vector db
3. Query
4. Search the vector database
5. Get meaningful output

Database used for embedding storage

- ChromaDB
- PgVector
- MongoDB
- LanceDB
- Cassandra
- Clickhouse
- and many more

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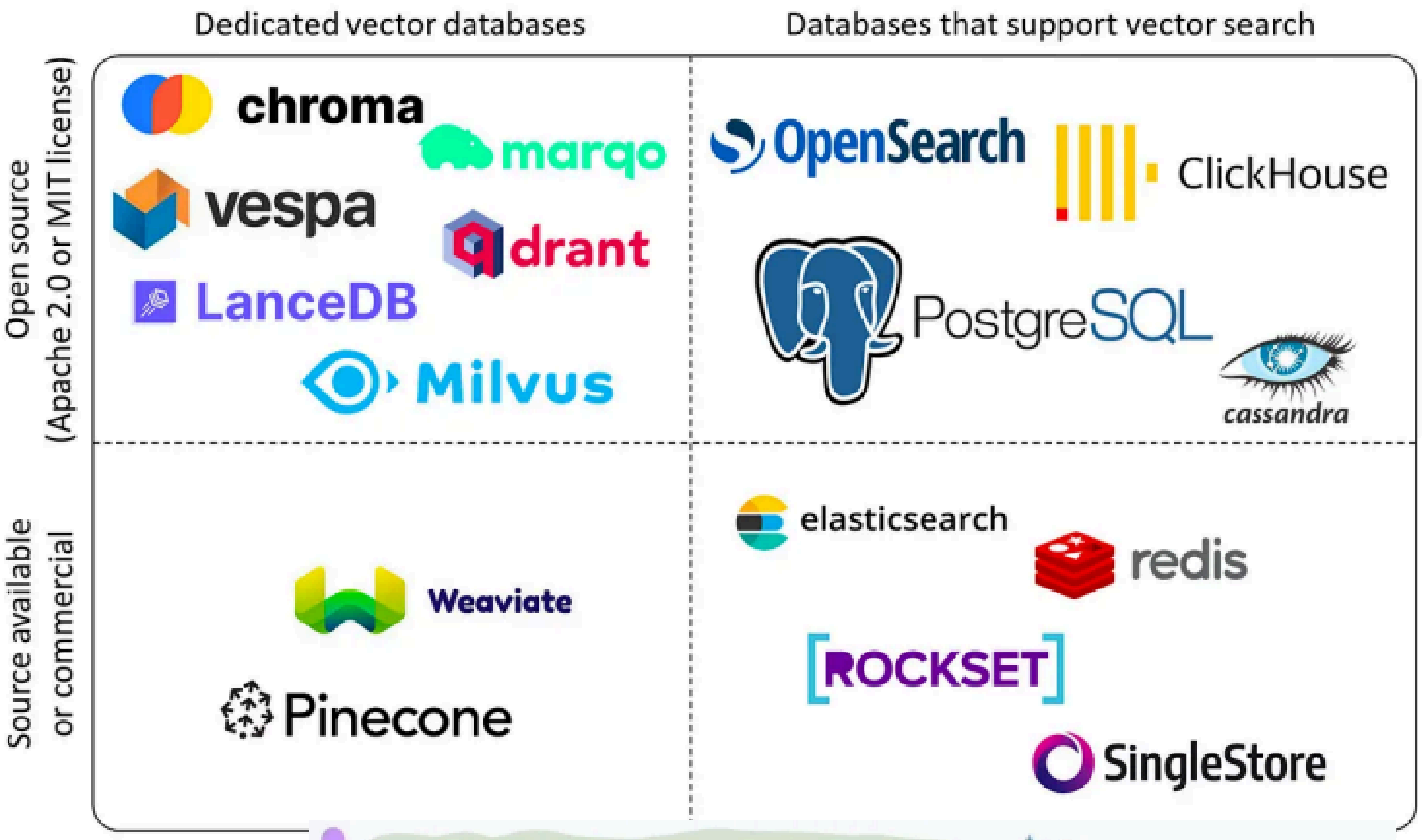
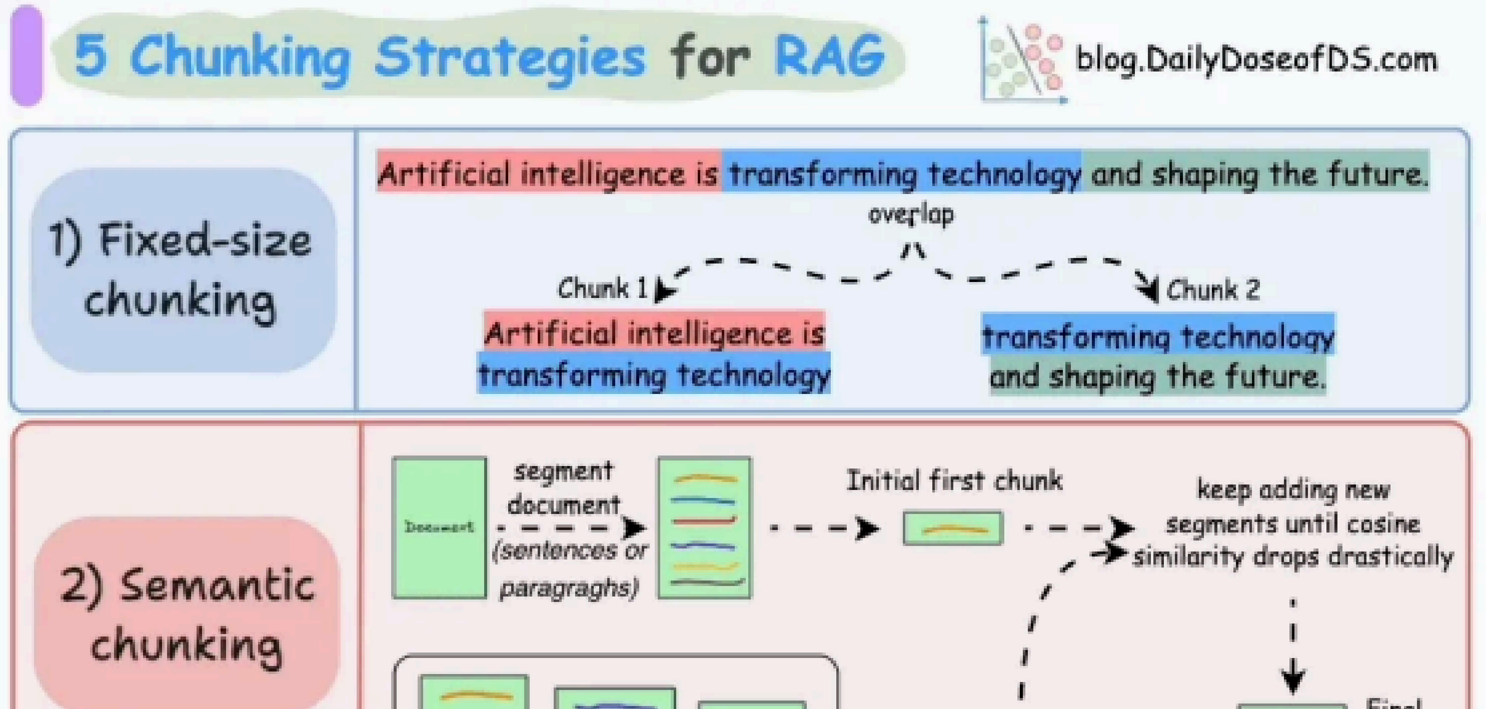


Chunking Approaches

- Fixed Size Chunking
- Semantic Chunking
- Recursive Chunking



Agentic Chunking

Searching

keyword search
vector search
hybrid search

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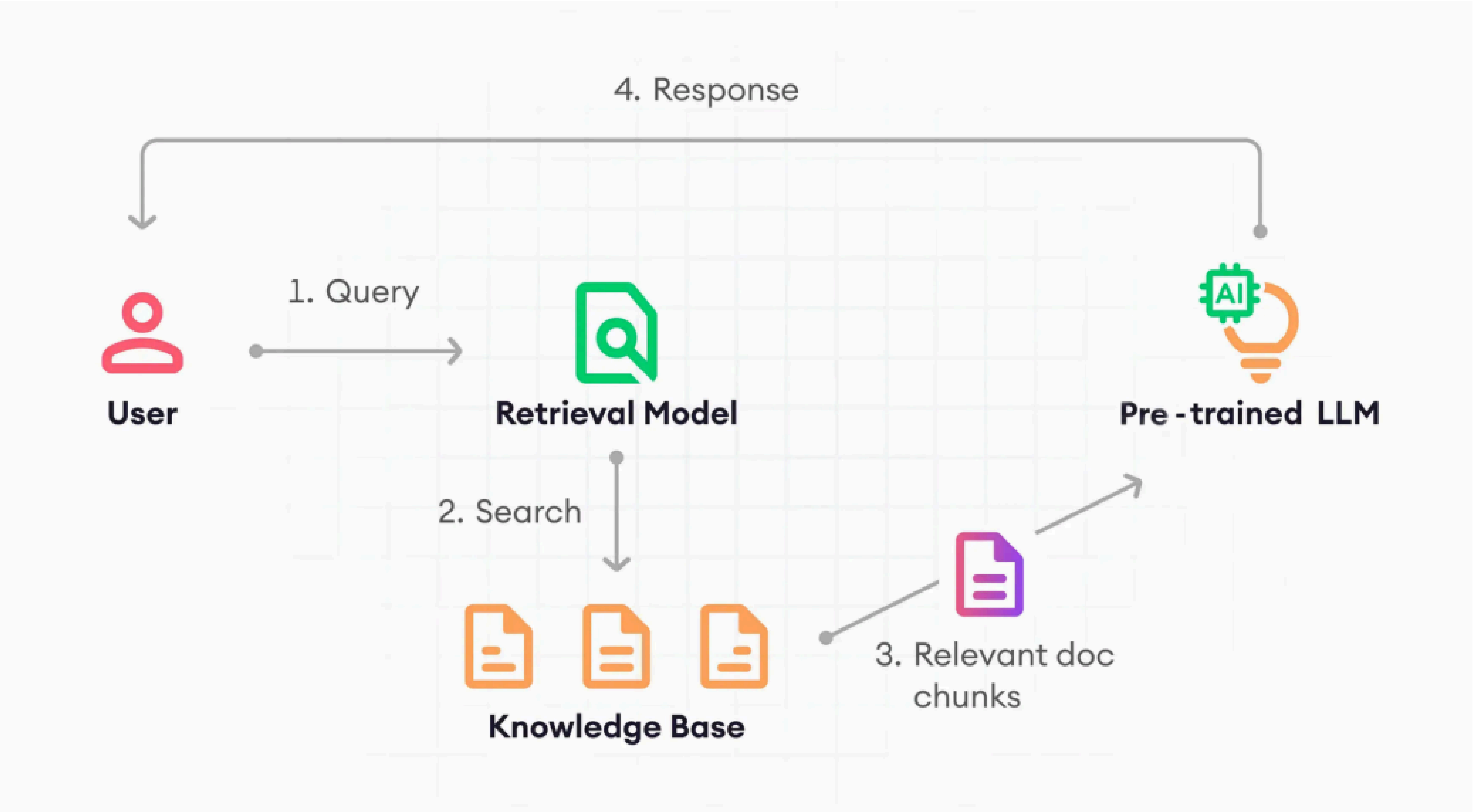
Working of Retrieval-Augmented Generation(RAG)



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Working of RAG

Working Mechanism:

1. Retrieval : searches and retrieves the relevant documents or data based on the user query.
2. Augmentation: Retrieved information is combined with the user query to provide context.
3. Generation : LLM model uses this augmented input to generate accurate response.

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Why use RAG ?

Benefits Of RAG

1. Prevent models from hallucinating
2. Allows model to cite sources
3. Expands the model use cases

4. Can stick to the internal knowledge base as required.

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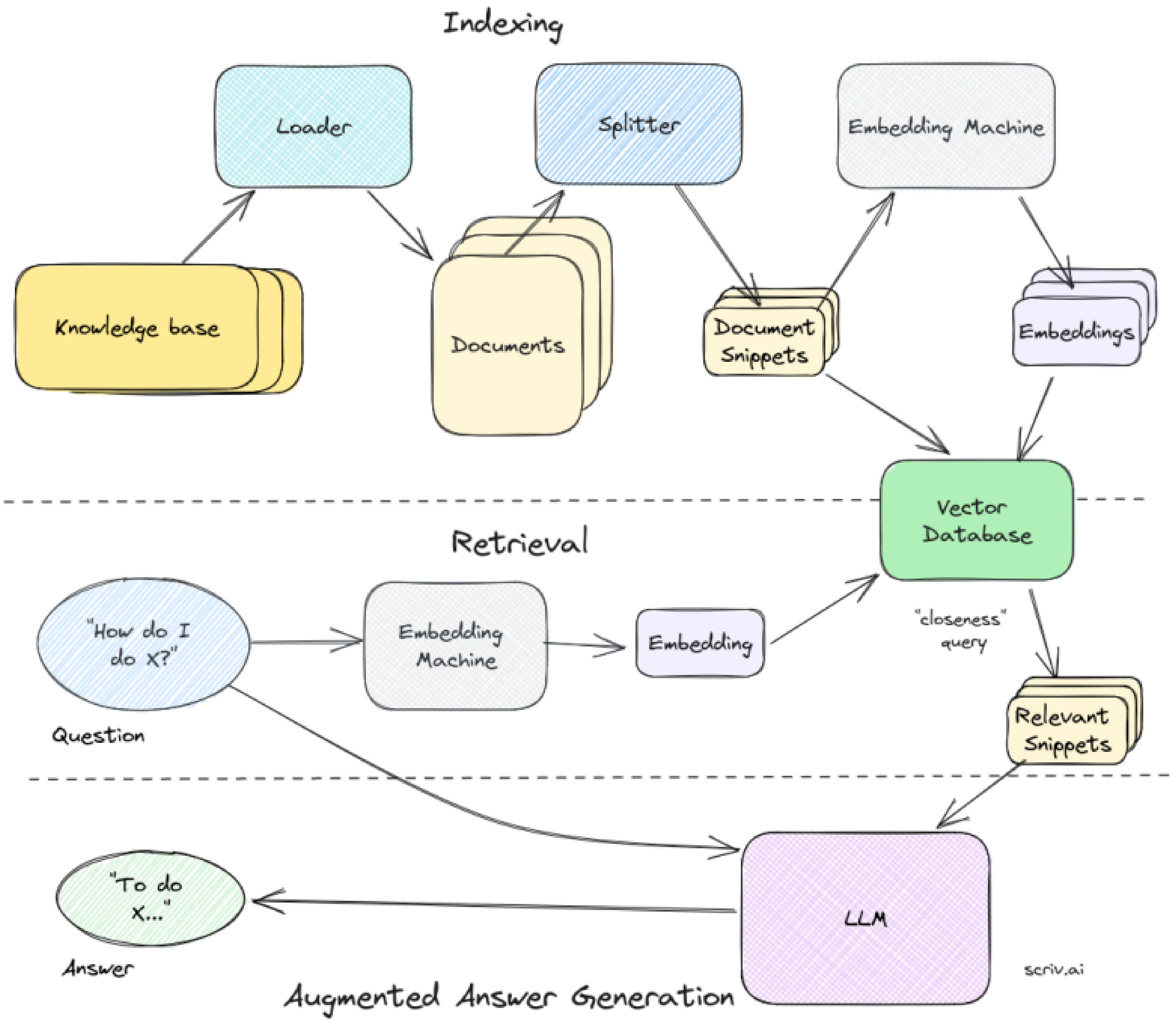
Use of RAG in AI Applications


Indexing
Loader
Splitter
Embedding Machine
Knowledge base
Documents
Document Snippets
Embeddings
Vector Database
Retrieval
"How do I do X?"
Question
Embedding Machine
Embedding
"closeness" query
Relevant Snippets
"To do X..."
Answer
LLM
scriv.ai
Augmented Answer Generation

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LLM Agent Frameworks

Some of the popular frameworks:

- Langchain's LangGraph
- Agno
- Autogen
- CrewAi

Why use these??

- Easy to switch LLM models
- Can easily add knowledge base
- Built-in Memory support
- Custom tool support, along with some built-in tools

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Let's Build Agentic AI Apps with Langchain

What we will do :

Set Up Environment with Langchain libraries and API keys

Create API Tools

Initialize Agent

Test the agent with and without tool call

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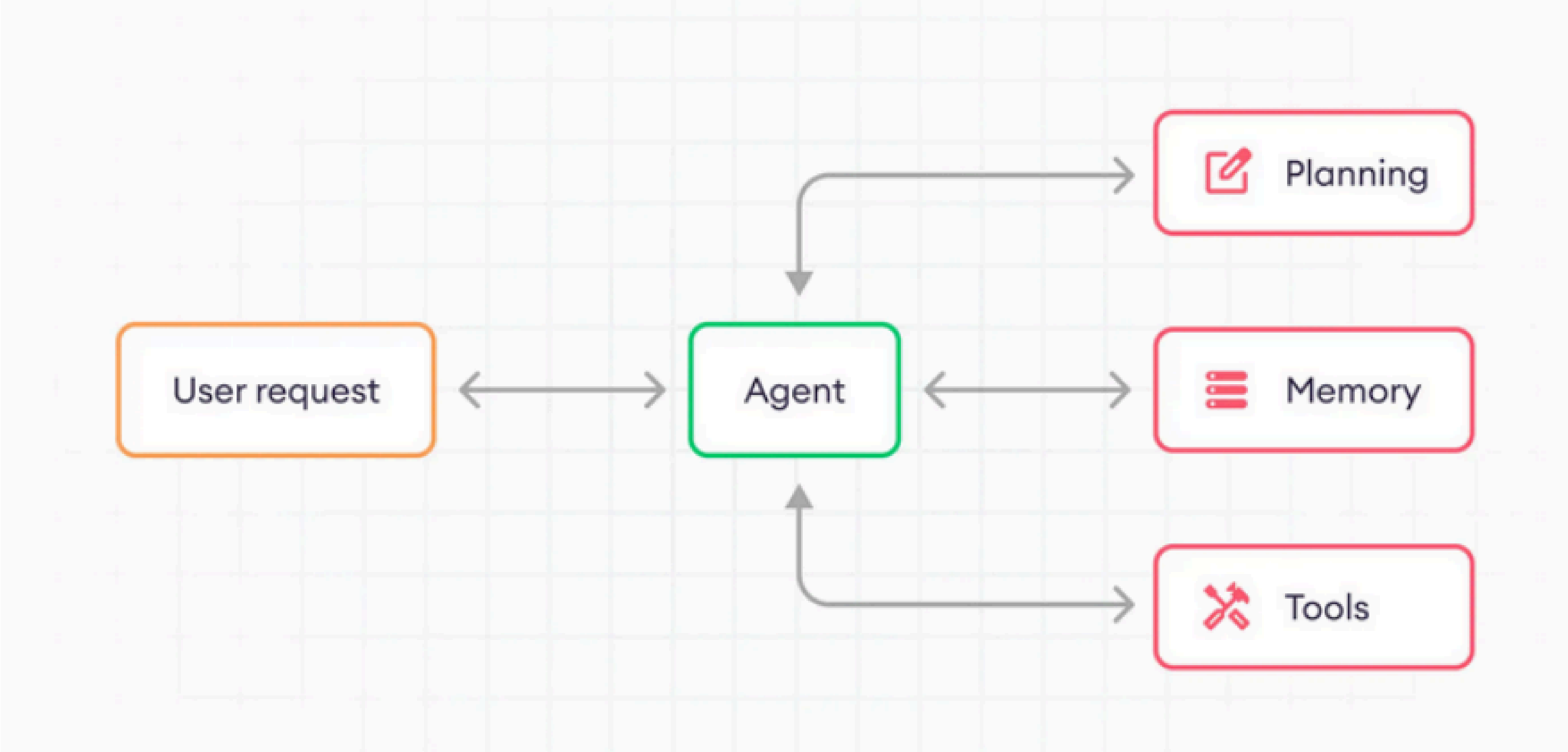
Single agent System

A system where one autonomous entity manages the entire workflow from perception to action.

Acts as a "central brain" that reasons, plans, and executes tasks independently.



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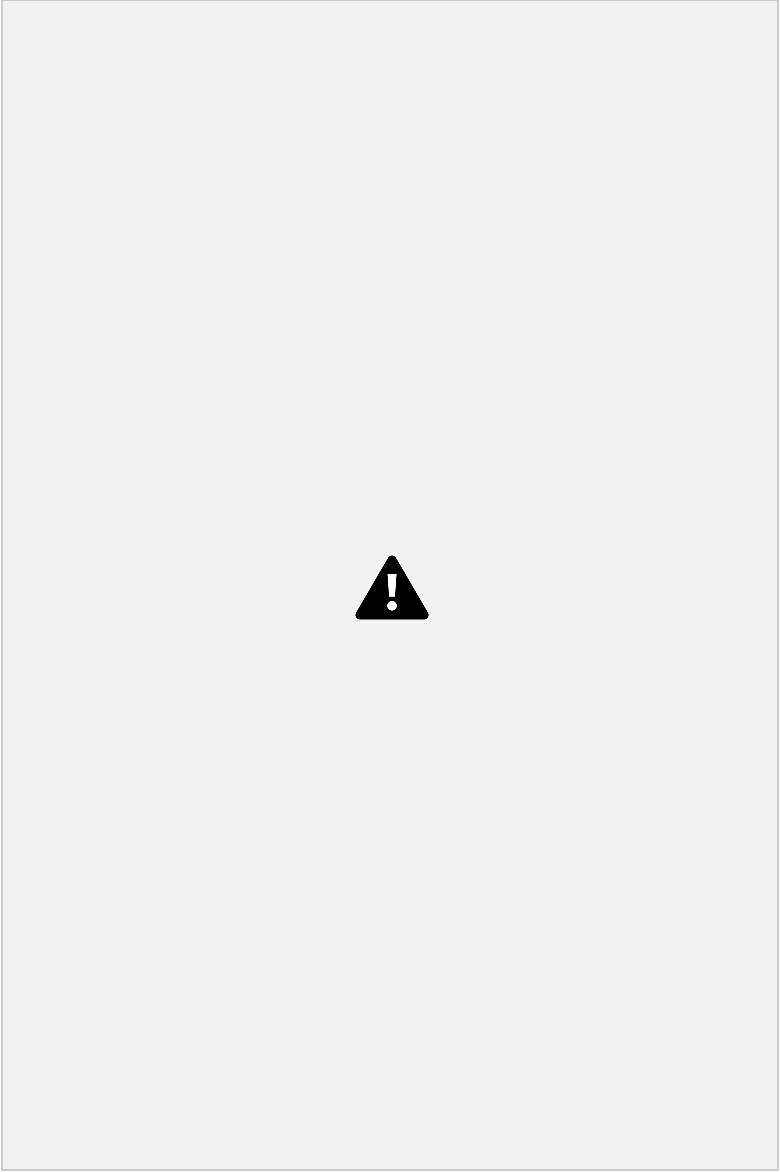
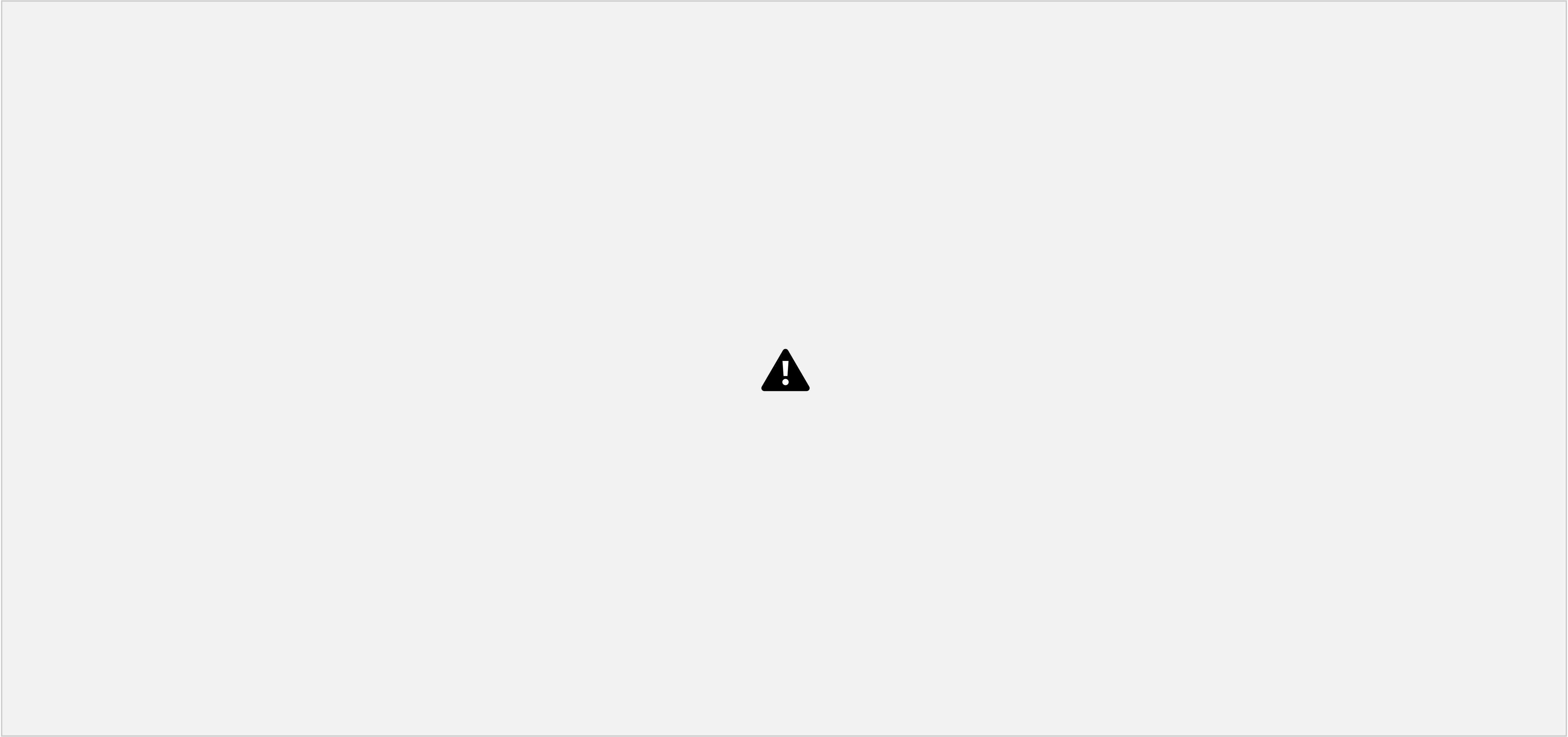
Multi-agent System

Combination of the agents to form a team
Imagine office hierarchy : under a project there will be multiple employee . And combination of them is Team.

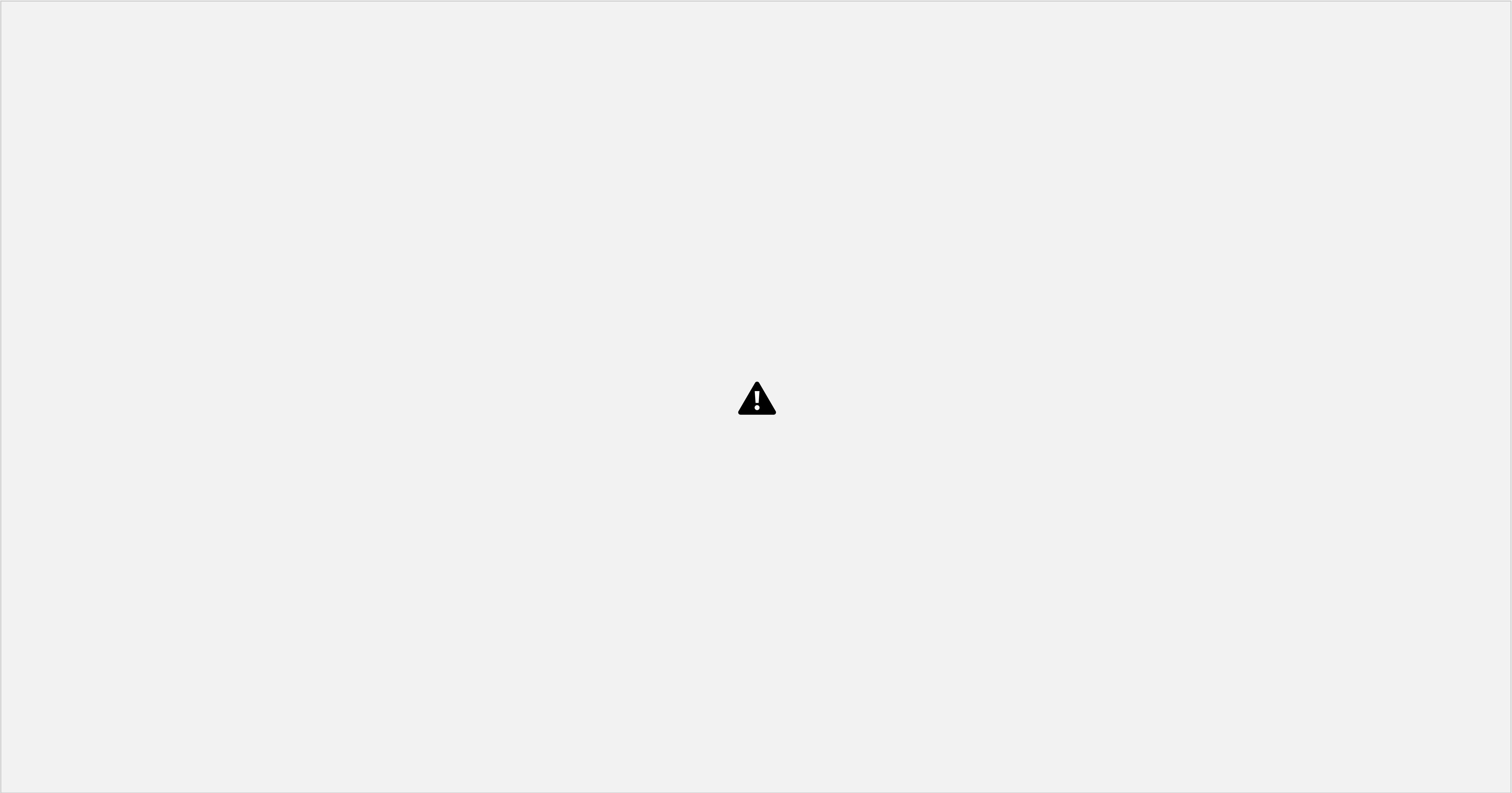
Same in this too ,Multiple agents with different roles are combined to form an team to complete certain task.





Guardrails

A programmable layer that sits between the User and the AI Agent to monitor inputs and outputs.
Acts as a "filter" to ensure the AI stays within legal, ethical, and technical



boundaries.

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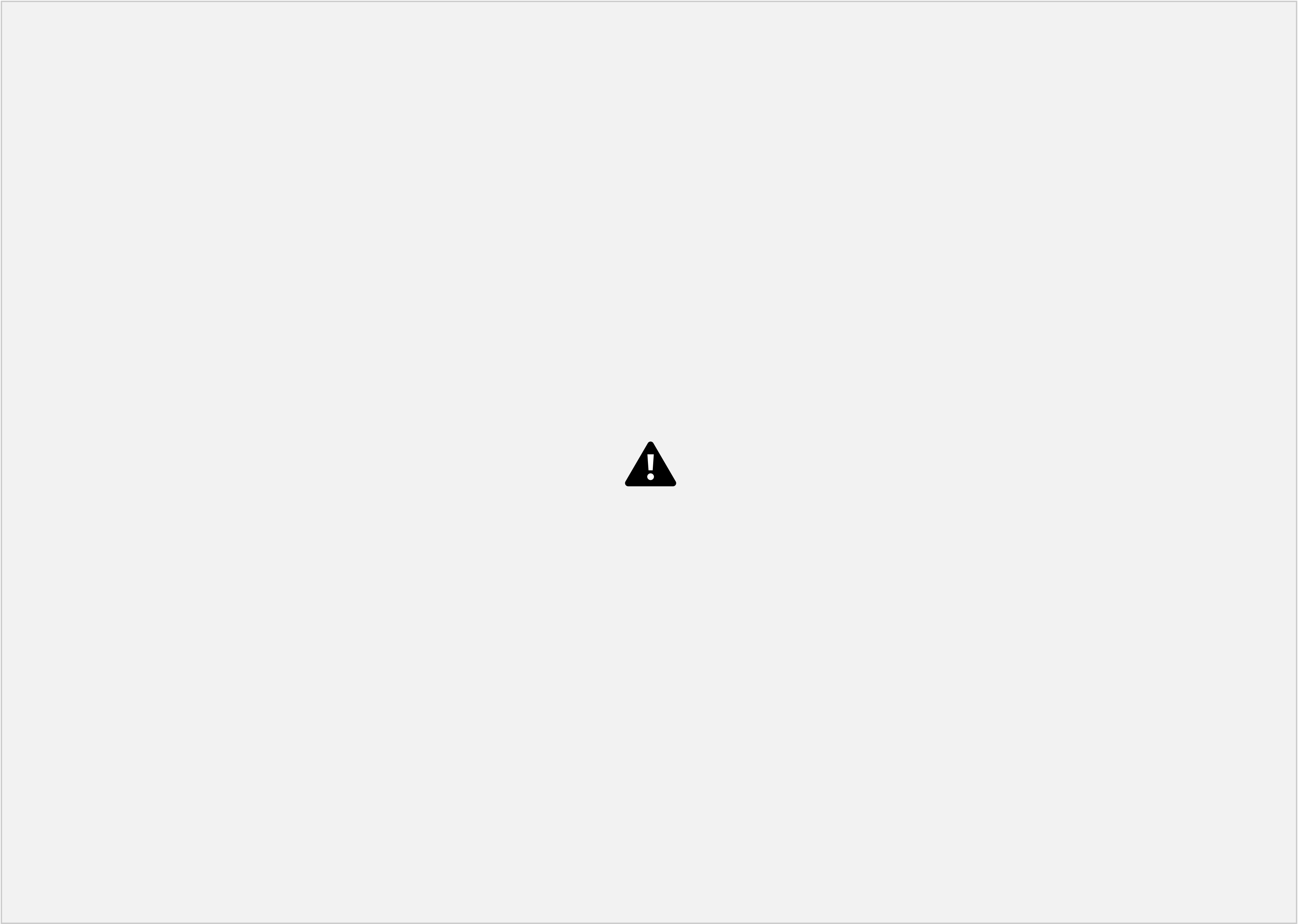
Human-in-the-loop

A model where human intervention is integrated into the AI's decision-making process.
Humans act as a checkpoint to approve, reject, or refine the agent's actions.



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Model context protocol (MCP)

An open standard that allows AI models to connect seamlessly to external data and tools.

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Task

Open task : create any agentic system which contains tools use, vector database , multi agents.

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Thank You

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